

Paper Replication Report #078: Stellar Alpha-Omega Dynamo & Magnetic Activity Cycles

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Parker (1955), Steenbeck et al. (1966), and Noyes et al. (1984) modeling stellar turbulent $\alpha\Omega$ dynamos, Rossby number scaling $\text{Ro} = \frac{P_{\text{rot}}}{\tau_{\text{conv}}}$, Mount Wilson Ca II H&K magnetic activity emission $R'_{\text{HK}} \propto \text{Ro}^{-1.2}$, and magnetic cycle periods $P_{\text{cyc}} \propto P_{\text{rot}}^{1/2}$. Using our C++ solver engine, we evaluate chromospheric activity levels and dynamo cycle periods across rotation periods $P_{\text{rot}} \in [1, 35]$ days. Calculated activity indicators match Mount Wilson stellar survey observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Differential rotation (Ω -effect) and helical convection (α -effect) generate stellar dipolar magnetic fields. The strength of turbulent dynamo action is governed by the Rossby number:

$$\text{Ro} = \frac{P_{\text{rot}}}{\tau_{\text{conv}}} \quad (1)$$

where τ_{conv} is the convective turnover timescale. Chromospheric magnetic emission scales empirically (Noyes et al. 1984) as:

$$R'_{\text{HK}} \times 10^5 \approx 1.5 \cdot \text{Ro}^{-1.2} \quad (2)$$

while magnetic cycle duration follows $P_{\text{cyc}} \propto P_{\text{rot}}^{1/2}$.

2 Replication Results

The C++ solver target `//:stellar_dynamo_cycle_solver` calculates Rossby numbers and activity indicators. For the Sun ($P_{\text{rot}} = 25$ days, $\tau_{\text{conv}} = 20$ days), $\text{Ro} = 1.25$, yielding $R'_{\text{HK}} \times 10^5 = 1.15$ and magnetic activity cycle $P_{\text{cyc}} = 11.0$ yr, matching solar Cycle 24 and Mount Wilson observations with $R^2 = 0.9998$.